AIE1901: AI Exploration I

Lab 3: Transfer Learning Guideline

Total marks: 100

Introduction

Transfer learning starts with a model trained on a **source task** and adapts it to a **target task**. In this lab, you will learn how to train a neural network and apply **fine-tuning**, then explore how data and model configurations affect transfer performance.

- **Part 1: Provided-data experiments (20 marks).** Continue the **Bananameter** banana classification example from class. The notebook walks you through training a neural network and fine-tuning it under different configurations. Submit a **PDF report**; no presentation is required.
- **Part 2: Your image project (80 marks).** Design a transfer problem using your own images and train your own source model. **If your project used audio or poses, download and use the image dataset provided on Blackboard in Lab 2 instead.** Submit **PowerPoint slides** and give a **5-minute presentation in English**. Each group may choose one or more members to present. All group members must attend and be prepared to answer questions. No separate Part 2 report is required.

AI assistance: You may use AI tools to assist with coding and other work. You must run the experiments and analyse the results yourselves, and understand what your submitted code does.

Getting Started

Use PyCharm, VS Code, or another IDE with your configured Python environment. Download and extract `AIE1901_Lab3.zip` from Blackboard, keeping the supplied folder structure:

```
AIE1901_Lab3/
|-- DATA/
|   |-- DataBanana2/<class folders>
|   |-- DataBanana1/<class folders>
|-- lab3/
|   |-- util.py
|   |-- requirements.txt
|   |-- 01_provided_data.ipynb
|   |-- 02_own_images.ipynb
```

Open the `lab3/` folder in your editor and follow the notebooks in numerical order. Step 1 trains the Part 1 source model; the notebooks create the model and result files when you run them. See **Before you run** in each notebook for package installation and running instructions.

Dates and Submission

Item	Deadline or arrangement
Part 1 Package	30 October 2026, 11:59 pm
Part 2 Package	1 November 2026, 11:59 pm. The package must include the PowerPoint slides for your presentation.
Part 2 presentation	2 November 2026, 5 minutes, in English, in ascending order of group number

Only one member needs to submit on behalf of the group. Submit the Part 1 ZIP file to the **Lab 3 Part 1 submission area** on Blackboard, and the Part 2 ZIP file to the **Lab 3 Part 2 submission area** on Blackboard. Replace `TeamNumber` with your actual team number.

Run the cells in each notebook from top to bottom and save both notebooks with their outputs before submission.

Part 1 Package

```
Lab3_Part1_TeamNumber.zip
|-- report.pdf
|-- 01_provided_data.ipynb       (executed, with outputs)
|-- outputs/part1/              (source_mlp.pt, CSV tables, plots,
                                 settings, histories and data manifests)
```

Part 2 Package

```
Lab3_Part2_TeamNumber.zip
|-- slides.pptx
|-- 02_own_images.ipynb         (executed, with outputs)
|-- outputs/part2/              (12-run results, summary/slide tables,
|                               error-bar plot, source/target curves,
|                               settings and trained models,
|                               including source_model.pt and
|                               selected_target_model.pt)
|-- my_images/
    |-- source/train/<class folders and images>
    |-- source/test/<class folders and images>
    |-- target/train/<class folders and images>
    |-- target/test/<class folders and images>
```

The provided Part 1 dataset does not need to be resubmitted. Your trained Part 1 source model is saved in `outputs/part1/` and included with your results. If you modify `util.py`, include your modified copy in the corresponding ZIP file and briefly explain the change. Use actual class folder names in Part 2.

At the beginning of the report and at the end of the slides, list every member's name, student ID, and **percentage contribution**. For each deliverable, contributions must sum to 100%.

Part 1: Provided-Data Experiments (20 marks)

Aims. Use **Bananameter** to practise image classification, understand neural network training and fine-tuning, and investigate how data and model configurations affect transfer performance.

Lab walkthrough. Follow `01_provided_data.ipynb`, Steps 1–4. The notebook explains the implementation and provides the experiment, table and plotting code.

Your Task:

Run the provided experiments and organise your PDF report into the four sections below. Complete each listed task using your measured results. Use **balanced accuracy** as the main comparison metric: the average of the accuracy within each class. Overall accuracy is saved for reference.

1. **Data and source model — Step 1 (2 marks).**
   - Describe the source and target classification tasks. Include the dataset names, classes, a training/test image-count table and representative images.
   - Report the source model's balanced accuracy on the source and target test sets before target training.
2. **Layer freezing — Step 2 (5 marks).**
   - Report the trainable parameter count and target training/test balanced accuracy for all four strategies in one table.
   - Include the strategy comparison plot and the training/test learning curves for all four strategies.
   - Compare the test results and identify the best-performing strategy in this experiment. Explain whether updating more parameters improved performance, using the results and learning curves as evidence.

   Results and evidence: 3 marks; interpretation: 2 marks.
3. **Target training size — Step 3 (6 marks).**
   - Report mean target test balanced accuracy and standard deviation (SD) across three runs for both methods at each of the six training sizes.
   - Include the summary table and plot mean balanced accuracy against training images per class, with error bars showing plus/minus one SD.
   - Compare transfer with no transfer at each size. Describe the trend, identify sizes where transfer helps or does not help, and explain one source of variability or one limitation.

   Results and evidence: 3 marks; interpretation: 3 marks.
4. **Covariate versus Semantic Shift — Step 4 (7 marks).**
   - Describe the classes and image conditions for the original reference and the two shifted tasks. State what changes and what stays fixed; include representative images.
   - Report the source-model reference table using balanced accuracy. Describe the camera change's effect before target training, explain the semantic N/A entry and explain the head choices.
   - Report mean target test balanced accuracy and SD across three runs for all five strategies within each shifted task. Include the summary table and comparison plot with error bars showing plus/minus one SD.
   - Compare freezing choices within each task and explain the results in terms of reusing learned features. Describe one limitation of comparing the two tasks, including differences in class difficulty.

   Design and results: 4 marks; interpretation: 3 marks.

The notebook contains the same report tasks. Label figures and tables clearly.

Part 2: Your Image Project (80 marks)

AI assistance: You may use AI tools to generate code. Understand its logic, run the experiments yourselves and analyse the actual results.

Choose Your Transfer Problem

Apply what you learned from Bananameter to your own images. Choose **one of two approaches**:

- **Semantic shift: change the task's classes.** For example, train on two classes, then transfer the hidden features to a three-class task using a new head.
- **Covariate shift: change image conditions while keeping the classes.** For example, train a source model using images taken under normal lighting, then fine-tune it using target training images taken under dim lighting. Evaluate the adapted model on separate target test images taken under the same dim-lighting condition. Keep the classes and their label meanings unchanged. You may reuse your Lab 2 conditions if they change image appearance, such as lighting or background, while keeping the classes and their label meanings unchanged.

Your Task:

Follow `02_own_images.ipynb`, Steps 5–7. **Modify the settings for your source and target tasks**, choose or modify an MLP, and train a source model. Explain the data split, model, frozen layers and head choices. Compare three freezing configurations with training from scratch, using **three seeds (42, 43, 44) per configuration: 12 target runs**. Keep the source model and data split fixed.

Use **target test balanced accuracy** as the main metric. Report each configuration's mean and standard deviation (SD), with error bars showing **plus/minus one SD across the three seeds**. The notebook supplies the result-table format and all repeat, summary and plotting code. Include its summary table and plot in your slides. Select the highest mean balanced accuracy, breaking ties by lower mean test loss. Save seed 42 of that configuration as the example model.

Report all outcomes, including cases where transfer does not help.

Component	Marks
Transfer question and data design (Step 5)	20
Model choice and source training (Step 6)	15
Transfer experiment, comparisons and interpretation (Step 7)	25
Slides and presentation (Step 8)	20
Part 2 total	80

Required Presentation Order

On **2 November 2026**, each group gives a **5-minute presentation in English** using the submitted PowerPoint slides. Each group may choose one or more presenters. All group members must attend and be prepared to answer questions. Follow this order:

1. **Introduce your dataset.** Show examples, classes or conditions, and the source/target training/test image counts.
2. **Introduce your transfer problem and design.** State the shift type, source-to-target changes and comparison design.
3. **Explain your experiments and results.** Present your source results and freezing/head choices. Show the four-configuration summary table and balanced-accuracy error-bar plot. Compare trans-

fer with no transfer and compare freezing settings using three-run means and SDs. Support your interpretation with source and selected target learning curves, and state one limitation.